EC200U&EG91xU Series

eSIM Application Note

LTE Standard Products

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About the Document

Revision History

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-	2024-01-18	Neo Jiang	Creation of the document
1.0	2024-08-09	Neo Jiang	First official release
1.1	2026-01-09	Farris Feng	1. Updated the declaration of AT command examples (Chapter 2.2). 2. Added contents related to the URC +QESIM: "add profile",<iccid> (Chapters 2.4.1 and 3.2).
1.2	2026-03-25	Kevin Xie	1. Added a note for AT+QESIM="ntf_get" and the related example (Chapter 2.3.10). 2. Updated the note for URC +QESIM: "add profile",<iccid> (Chapters 2.4.1 and 3.2).

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1 Introduction

The embedded SIM (also called eSIM or eUICC, referred to as eSIM hereinafter) card is a new secure element compliant with GSMA specifications that is designed to remotely manage multiple mobile network operator subscriptions. Available in various form factors, either plugged-in or soldered, the eSIM is easy to integrate in any device.

This document outlines the eSIM AT commands of EC200U series [1] and EG91xU family (comprising EG912U series and EG915U series) modules. These commands are used to download encrypted profiles to eSIM and provide a local management interface for managing eSIM profiles.

[1] EC200U-CN excluded since the model does not support eSIM by default. Please contact Quectel Technical Support for eSIM demand on EC200U-CN.

2 eSIM AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	**AT+<cmd>=?**	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	**AT+<cmd>?**	Check the current parameter value of the corresponding command.
Write Command	**AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]**	Set user-definable parameter value.
Execution Command	**AT+<cmd>**	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence. The URLs, domain names, IP addresses, usernames/accounts, and passwords (if any) in the AT command examples are provided for illustrative and explanatory purposes only, and they should be modified to reflect your actual usage and specific needs.

2.3. Local Management Commands

2.3.1. AT+QESIM="list" List Profiles

This command lists profiles currently installed in the eSIM.

AT+QESIM="list" List Profiles	
Write Command **AT+QESIM="list"[,<mode>]**	Response **+QESIM: "list",<result>** **[<ICCID>,<status>,<nickname>,<profile_class>,<profile_name>,<provider>** **[...]]** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	-

Parameter

<mode>	Integer type. Profile list mode. 0 Only list operational profiles in eSIM 1 List all profiles in eSIM (including test, provisioning and operational profiles)
<ICCID>	String type. Profile ICCID.
<status>	Integer type. Profile status.

	0 Disabled 1 Enabled
<nickname>	String type. Profile nickname. The value can be null.
<profile_class>	Integer type. Profile class. 0 Test profile 1 Provisioning profile 2 Operational profile
<profile_name>	String type. Profile name.
<provider>	String type. Service provider name.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="list",1
+QESIM: "list",0
"8900000000000000060F",0,"",0,"Rohde&Schwarz","Rohde&Schwarz"
"89320420000017901669",1,"",2,"Quectel_eSIM_04","BICS"

OK
```

2.3.2. AT+QESIM="enable" Enable a Profile

This command enables a specified profile currently installed in the eSIM.

AT+QESIM="enable" Enable a Profile	
Write Command **AT+QESIM="enable",<ICCID>**	Response **+QESIM: "enable",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	This command takes effect immediately. The configuration is saved in eSIM.

Parameter

<ICCID>	String type. Profile ICCID.

<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="enable","89320420000000199885F"
+QESIM: "enable",0

OK
```

2.3.3. AT+QESIM="disable" Disable a Profile

This command disables the currently enabled profile in the eSIM.

AT+QESIM="disable" Disable a Profile	
Write Command **AT+QESIM="disable",<ICCID>**	Response **+QESIM: "disable",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	This command takes effect immediately. The configuration is saved in eSIM.

Parameter

<ICCID>	String type. Profile ICCID.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="disable","89320420000000199885F"
+QESIM: "disable",0

OK
```

2.3.4. AT+QESIM="delete" Delete a Profile

This command deletes a profile installed in eSIM. Only disabled profiles can be deleted. When the profile is deleted, the module tries to send the notification generated by the deletion to the SM-DP+ server.

If notification sending is successful, the module deletes this notification from eSIM. If notification sending fails, the module saves the notification to eSIM. In this case, check the network environment first, and then execute **AT+QESIM="ntf_report",<seq>** to resend the notification.

AT+QESIM="delete" Delete a Profile	
Write Command **AT+QESIM="delete",<ICCID>**	Response **+QESIM: "delete",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	60000 ms
Characteristics	-

Parameter

<ICCID>	String type. Profile ICCID.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="delete","89320420000000199885F"
+QESIM: "delete",0

OK
```

2.3.5. AT+QESIM="nickname" Set Profile Nickname

This command sets the nickname of a specified profile.

AT+QESIM="nickname" Set Profile Nickname	
Write Command **AT+QESIM="nickname",<nickname>,<**	Response **+QESIM: "nickname",<result>**

ICCID>	**OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	The command takes effect immediately. The configuration is saved in eSIM.

Parameter

<nickname>	String type. Profile nickname. Maximum length: 32 characters.
<ICCID>	String type. Profile ICCID.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="nickname","simcad_test","89320420000000199885F"
+QESIM: "nickname",0

OK
```

2.3.6. AT+QESIM="eid" Query eSIM ID

This command queries the eSIM ID of the module.

AT+QESIM="eid" Query eSIM ID	
Write Command **AT+QESIM="eid"**	Response **+QESIM: "eid",<result>[,<eid>]** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	-

Parameter

<eid>	String type. eSIM ID.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="eid"
+QESIM: "eid",0,"89049032000001000000092092075457"

OK
```

2.3.7. AT+QESIM="ntf_list" List Notifications

This command lists notifications stored in eSIM.

AT+QESIM="ntf_list" List Notifications	
Write Command **AT+QESIM="ntf_list",<type>**	Response **+QESIM: "ntf_list",<result>** **[<ntf_event>,<seq>,<ICCID>** **[…]]** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	-

Parameter

<type>	Integer type. Event type. 0 All events 1 Enable event 2 Disable event 3 Delete event 4 Installation event
<ntf_event>	Integer type. Profile management operation. 1920 Installation notification

	1600	Enable notification
	1312	Disable notification
	1040	Delete notification
<seq>	Integer type. Notification serial number.	
<ICCID>	String type. Profile ICCID.	
<result>	Integer type. Result code. See ***Chapter 4*** for details.	
<err>	Integer type. Error code. See ***document [1]*** for details.	

Example

```
AT+QESIM="ntf_list",0
+QESIM: "ntf_list",0
1920,178,"89320420000016766683"
1920,184,"89320420000016766683"
1920,244,"89860081192045066333"
1312,407,"89320420000016740225"

OK
```

2.3.8. AT+QESIM="ntf_delete" Delete a Notification

This command deletes a notification stored in eSIM.

AT+QESIM="ntf_delete" Delete a Notification	
Write Command **AT+QESIM="ntf_delete",<seq>**	Response **+QESIM: "ntf_delete",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	-

Parameter

<seq>	Integer type. Notification serial number.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="ntf_delete",412
+QESIM: "ntf_delete",0

OK
```

2.3.9. AT+QESIM="ntf_report" Send Notification to SM-DP+

This command sends a notification stored in eSIM to the corresponding SM-DP+. If the notification is sent successfully, the notification is automatically deleted from eSIM.

AT+QESIM="ntf_report" Send Notification to SM-DP+	
Write Command **AT+QESIM="ntf_report",<seq>**	Response **+QESIM: "ntf_report",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	60000 ms
Characteristics	-

Parameter

<seq>	Integer type. Notification serial number.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="ntf_report",412
+QESIM: "ntf_report",0

OK
```

2.3.10. AT+QESIM="ntf_get" Query Notification Details

This command queries the details of a notification stored in eSIM.

AT+QESIM="ntf_get" Query Notification Details	
Write Command **AT+QESIM="ntf_get",<seq>**	Response **+QESIM: "ntf_get",<result>[,<data>]** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	-

Parameter

<seq>	Integer type. Notification serial number.
<data>	String type. Details of the notification.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

NOTE

1. **<data>** is omitted when no notifications are stored in the eSIM.
2. **<data>** is null if the eSIM stores other notifications but not the one currently being queried.

Example

```
AT+QESIM="ntf_get",418
+QESIM: "ntf_get",0,"MIIFVL8vO4ACAaKBAgZADCV0aGFsZXMzLnByb2Qub25kZW1hbmRjb25uZ
WN0aXZpdHkuY29tWgqYI0ACAABhZ1aRXzdAiLv7VLSNwqgh1P65rx6ARDX/smV1UPf9lJiWAMCf
5fLQ0u4Drcl2Ob+rvD/Swy8r5zfSi601X//u3cxTjS+P+zCCAgUwggGsoAMCAQICDgRj5NPKyMMfCD
vXmZqCMAoGCCqGSM49BAMCMGIxCzAJBgNVBAYTAlVLMRIwEAYDVQQHDAlDYW1icmlkZ2Ux
DjAMBgNVBAoMBUtpZ2VuMRQwEgYDVQQLDAtFbmdpbmVlcmluZzEZMBcGA1UEAwwQRVVNQ
29uc3VtZXJER0R1YjAgFw0yMjEwMTExMDI3MzNaGA8yMTIyMTAxMTEwMjczM1owOzEOMAwGA
1UECgwFS2lnZW4xKTAnBgNVBAUTIDg5MDQ0MDQ1MTI2NzI3NDk0ODAwMDAwMDAwMDM5NT
U0MFkwEwYHKoZIzj0CAQYIKoZIzj0DAQcDQgAEoEQTG3fwfOCOwj6dN3bm0qs3SRRaVE+4sdX8
orlO4h9f7Aj77/TVx1rCdHFU5So5doUV347DceQfyhnk41/8E6NrMGkwHQYDVR0OBBYEFBT91Hv2e
7PLOD413+IlL4VKhzRyMB8GA1UdIwQYMBaAFBiaCFOv1JT2sHmaC2fw9dDBKeP2MA4GA1UdDw
```

```
EB/wQEAwIHgDAXBgNVHSABAf8EDTALMAkGB2eBEgECAQEwCgYIKoZIzj0EAwIDRwAwRAIgUf
4aAz+HCiSQ3GsN+B3FTyZWPZySz3s1rROqfEL9yQsCIBkpxRSvNnUK/ey6XDXqFTJXw8IKnUBH+
vbq83+fFvnoMIICxjCCAm2gAwIBAgIQWs6PmNyGSUBKva1IfiBQkzAKBggqhkjOPQQDAjBEMRgw
FgYDVQQKEw9HU00gQXNzb2NpYXRpb24xKDAmBgNVBAMTH0dTTSBBc3NvY2lhdGlvbiAtIFJTU
DIgUm9vdCBDSTEwHhcNMjIxMDA1MDAwMDAwWhcNNDkxMjMxMjM1OTU5WjBiMQswCQYDVQ
QGEwJVSzESMBAGA1UEBwwJQ2FtYnJpZGdlMQ4wDAYDVQQKDAVLaWdlbjEUMBIGA1UECww
LRW5naW5lZXJpbmcxGTAXBgNVBAMMEEVVTUNvbnN1bWVyREdEdWIwWTATBgcqhkjOPQIBB
ggqhkjOPQMBBwNCAATG+X7qL3aK0XuoYpqwBOz3kP9DV/IIG+FcIxn3LsIm+aJgu7XHNRYGI3d7
gptZyfZfvJo9eFQQ2Jhvwk74pDtbo4IBITCCAR0wEgYDVR0TAQH/BAgwBgEB/wIBADAXBgNVHSA
BAf8EDTALMAkGB2eBEgECAQIwTQYDVR0fBEYwRDBCoECgPoY8aHR0cDovL2dzbWEtY3JsLnN
5bWF1dGguY29tL29mZmxpbmVjYS9nc21hLXJzcDltcm9vdC1jaTEuY3JsMA4GA1UdDwEB/wQEA
wIBBjA3BgNVHR4BAf8ELTAroCkwJ6QlMCMxDjAMBgNVBAoTBUtpZ2VuMREwDwYDVQQFEwg4
OTA0NDA0NTAWBgNVHREEDzANiAsrBgEEAYO7EwECAjAdBgNVHQ4EFgQUGJolU6/UlPaweZoL
Z/D10MEp4/YwHwYDVR0jBBgwFoAUgTcPUSXQsdQl1MOyMubSXnIb6/swCgYIKoZIzj0EAwIDRwA
wRAIgEGYYqo+9pfQpkoH043EiBObp87CV8ZuOUu2mH35KTbECIDX4LQ40bi6DbH8YRDkqOS5D2
976ROC/IhUI6JTIk4PR"

OK
AT+QESIM="ntf_list",0
+QESIM: "ntf_list",0
1920,609,"89320420000017901776"
1600,610,"89320420000017901776"
1312,611,"89320420000017901776"

OK
AT+QESIM="ntf_get",1
+QESIM: "ntf_get",0,""     //<data> is null when the queried notification is not stored in the eSIM.

OK
```

2.3.11. AT+QESIM="fallback" Enable Fallback Profile

This command enables the Fallback Profile (configured by the Fallback Attribute) when the current profile becomes unusable, and sets ReturnFromFallback Attribute to the currently enabled profile. You can set a profile with the Fallback Attribute through eIM only, and only one profile can be set.

AT+QESIM="fallback" Enable Fallback Profile	
Write Command **AT+QESIM="fallback"**	Response **+QESIM: "fallback",<result>** **OK** If there is any error: **ERROR**

	Or **+CME ERROR: <err>**
Maximum Response Time	600 ms
Characteristics	This command takes effect immediately. The configuration is saved in eSIM.

Parameter

<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="list",1
+QESIM: "list",0
"89320420000016766444",0,"",2,"BICS","BICS"      //89320420000016766444 is the Fallback Profile
"89320420000016767194",1,"",2,"BICS","BICS"      //89320420000016767194 is the currently enabled operational profile

OK
AT+QESIM="fallback"      //Enable the Fallback Profile and set ReturnFromFallback Attribute to the currently enabled profile
+QESIM: "fallback",0

OK
AT+QESIM="list",1
+QESIM: "list",0
"89320420000016766444",1,"",2,"BICS","BICS"      //The Fallback Profile is enabled
"89320420000016767194",0,"",2,"BICS","BICS"      //89320420000016767194 is disabled and ReturnFromFallback Attribute is set

OK
```

2.3.12. AT+QESIM="fallback_return" Enable ReturnFromFallback Profile

This command disables the currently enabled Fallback Profile and re-enables the ReturnFromFallback Profile. If the command is executed successfully, the ReturnFromFallback Attribute is cleared.

AT+QESIM="fallback_return" Enable ReturnFromFallback Profile	
Write Command **AT+QESIM="fallback_return"**	Response **+QESIM: "fallback_return",<result>**

	OK If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	600 ms
Characteristics	This command takes effect immediately. The configuration is saved in eSIM.

Parameter

<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="list",1
+QESIM: "list",0
"89320420000016766444",1,"",2,"BICS","BICS"
"89320420000016767194",0,"",2,"BICS","BICS" //89320420000016767194 is the ReturnFromFallback Profile

OK
AT+QESIM="fallback_return"                  //Enable the ReturnFromFallback Profile and clear its ReturnFromFallback Attribute
+QESIM: "fallback_return",0                 //The ReturnFromFallback Attribute is cleared

OK
AT+QESIM="list",1
+QESIM: "list",0
"89320420000016766444",0,"",2,"BICS","BICS"  //89320420000016766444 is disabled
"89320420000016767194",1,"",2,"BICS","BICS"  //89320420000016767194 is enabled with its ReturnFromFallback Attribute cleared

OK
```

2.4. Profile Download Commands

2.4.1. AT+QESIM="download" Start Profile Download

This command starts the profile download. After the profile download and installation are completed, the module reports the URC **+QESIM: "add profile",<iccid>** and **+QESIM: "download",<result>[,<errinfo1>,<errinfo2>]** to indicate the profile download and installation result.

This command supports two profile download modes (see ***Chapter 2.4.3*** for details):

- Internal network mode;
- External network mode, in which this command has to be used with **AT+QESIM="trans"** in ***Chapter 2.4.2*** to complete the profile download and installation. See ***Figure 3*** for the profile download and installation flowchart.

AT+QESIM="download" Start Profile Download	
Write Command **AT+QESIM="download",<nw_mode>,<activation_code>[,<confirmation_code>]**	Response **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	-

Parameter

<nw_mode>	Integer type. Profile download mode. 0 Internal network mode 1 External network mode. If the command is executed successfully, the following URC is reported: If **<result>** is 0: **+QESIM: "trans",<result>,<URL>,<payloadlen>,<payload>** If **<result>** is not 0: **+QESIM: "trans",<result>[,<errinfo1>,<errinfo2>]**
<activation_code>	String type. Activation code, which is used to request the download and installation of a profile. Activation codes are typically obtained from the network service provider or from Quectel directly, and they are structured as illustrated in ***Table 2***.
<confirmation_code>	String type. Confirmation code, which is used to validate the request to download and install a profile.
<err>	Integer type. Error code. See ***document [1]*** for details.

Table 2: Activation Code Structure

Name	Description	MOC
AC_Format	Format of the activation code. It is "1" for this format of the activation code and any subsequent backward compatible format.	M
Delimiter	"$"	M
SM-DP+ Address	FQDN of the SM-DP+ (e.g.,SMDP.GSMA.COM) restricted to the alphanumeric mode character set defined in ***Table 5*** of *ISO/IEC 18004* excluding "$".	M
Delimiter	"$"	M
AC_Token	Matching ID as described in ***Section 4.1.1*** of *SGP.22*.	M
Delimiter	Present and set to "$" if any of the following optional parameters is present.	C
SM-DP+ OID	SM-DP+ OID in the *CERT.DPauth.SIG*	O
Delimiter	Present and set to "$" if any of the following optional parameters is present.	C
Confirmation Code Required Flag	Present and set to "1", if confirmation code is required; otherwise it is absent.	O
Delimiter	Present and set to "$" if any of the following optional parameters is present.	C
CI Public Key Indicator	If present, it specifies an eSIM CA RootCA public key as specified in ***Section 4.1.2*** of *SGP.22*.	O
Delimiter	Present and set to "$" if any of the following optional parameters is present.	C
Delete Notification for Device Change	If present in an activation code, it specifies the delete notification for device change. See ***Section 4.1.3*** in *SGP.22*. This parameter is present only in an activation code transferred (directly or indirectly) from the old to the new device in the context of a device change.	O

Example

```
AT+QESIM="download",0,"LPA:1$demo-consumer.validspereachdpplus.com$1BINSTP8J0EFCW
LE4JY4S0YGO7II40ZXSQESI06THYWPLZ-W65CXYWK5-KPZJCZ"
OK
```

NOTE

1. When provided in a QR code according to *ISO/IEC 18004*, the activation code is prefixed with "LPA:".
2. The above URCs are detailed in ***Chapter 3***.
3. The URC **+QESIM: "add profile",<iccid>** applies only to the EC200U series and EG91xU family modules that use R01, R03 and R06 firmware versions. R01, R03 and R06 firmware refers to the firmware whose version number contains “R01”, “R03”, or “R06” as a segment, such as

EC200UxxxxR01AxxMxx, EC200UxxxxR03AxxMxx, and EC200UxxxxR06AxxMxx. For details, please contact Quectel Technical Support.

2.4.2. AT+QESIM="trans" Forward SM-DP+ Message to Module

This command forwards the message returned by SM-DP+ to the module during the profile download process when an external network is used. If the command is executed successfully, either of the following URCs is reported:

- If **<result>** is 0: **+QESIM: "trans",<result>,<URL>,<payloadlen>,<payload>**;
- If **<result>** is not 0: **+QESIM: "trans",<result>[,<errinfo1>,<errinfo2>]**.

AT+QESIM="trans" Forward SM-DP+ Message to Module	
Write Command **AT+QESIM="trans",<totallen>,<state>,<seq>,<len>,<data>**	Response **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	6000 ms
Characteristics	-

Parameter

<totallen>	Integer type. Total length of the SM-DP+ message after it is converted into a hexadecimal string. Range: 1–65534, must be an even number. Unit: byte. If **<totallen>** is greater than **<len>**, the SM-DP+ message must be divided into multiple data packets for forwarding.
<state>	Integer type. Data block state. 0 Last packet of SM-DP+ message 1 Not last packet of SM-DP+ message
<seq>	Integer type. Serial number. Range: 0–255.
<len>	Integer type. Length of data to be sent. Range: 1–1400, must be an even number. Unit: byte.
<data>	Hexadecimal string type. SM-DP+ message data to be forwarded to the module.
<err>	Integer type. Error code. See ***document [1]*** for details.

NOTE

The above URCs are detailed in ***Chapter 3***.

Example

```
AT+QESIM="trans",2878,1,0,1400,"485454502F312E3120323030204F4B0D0A436F6E74656E742D5
47970653A206170706C69636174696F6E2F6A736F6E0D0A446174653A205468752C203235204E6F7
6203230323120303333A34373A333820474D540D0A582D41646D696E2D50726F746F636F6C3A2067
736D612F7273702F76322E322E320D0A0D0A7B2268656164657223A7B2266756E6374696F6E457
865637574696F6E537461747573223A7B2273746174757322 3A2245786563757465642D5375636365
7373227D7D2C227472616E73616374696F6E4964223A22323032313131323530333437333845373 94
4424232353842394539334139434122 2C2273657276657253696 76E656431223A224D4643414543416
845535544527A6A6E6E62736C693536547163714245485 8332F7A31433634573241414378417 2456E
63586D44474752774D69356A5A4331305A584E304C6D56314C6D74705A3256754C6D4E766259515
15057324E586A36532F4D71713144424D427255546A513D3D222C2273657276657253696 76E61747
5726531223A22587A6441784835 4C336C4351747359344566 6E6475644E4574366946395172474968
53473734356B664A7A734D4B326A704B4C4B476C6A326B3343363450494A754932414772674A6D
67624F70537044544A4647703274634351 3D3D222C2265756963634369504B4964546F42655573656
4223A224242526D5768517A316E77614C4632347453795766784367563770637367 3D3D222C227365
72766572436572746966696361746522 3A224D494943696A434341692B6741774942416749514F4531
6545464432416C625033537A4868456934505441 4B42676771686B6A4F50515144416A426F4D52307
747775944565151 4B457852546557316862 6E526C5979424462334A7762334A6864476C76626A4648
4D4555474131554541784D2B55336C74595735305A574D6751323979634739"
OK
AT+QESIM="trans",2878,1,1,1400,"795958527062323467556C4E514946526C63335167556D397664
434244515341744945 5A76636942555A584E3049464231636E42766332567A4945397562486B77486
8634E4D6A45774D6A497A4D4441774D4441775768634E4D6A51774D6A497A4D6A4D314F5455355
76A41384D534577487759445651514444426 86B6344457559325174644756 7A6443356C6453357261
57646C6269356A62323078467A415642674E5642416F4D446B744A5230564F4943685653796B6754
4652454D466B77457759484B6F5A497A6A3043415159494B6F5A497A6A3044415163445167414544
463949345A656F6A2F4973694257795870727 46B61316D5472674A452F343679517A56632F684E647
377755146476C734D317A347879435A4C356441662B59476E584A5777756335565968786641416959
536453714F42356A4342347A414F42674E5648513842416638454241 4D4342344177485159445652304
F4242594546447 15A486443515A32584B68374973525350724B782B494530384E4D426B474131556
4455151534D4243494469734741515142673773546956494241674D424D4263474131556449414542 2
F77514E4D417377435159485A3445534151494242444264426 74E5648523845566A42554D464B6755
4B424F686B786F644852774F693876634774704C574E796243357A655731686458526F4C6D4E7662
53396A595639684D32526A4D6D557A5A6D56684E7A63774F4745784D574D344F446B7A4F445A6B
4F57517A595463325A69394D5958526C63335244556B777559334A734D4238474131556449775159 4
D42614146475A614644505766426F73586269314C4A5A2F454B4258756C79794D416F47434371475
34D343942414D4341306B414D4559434951434D5248716666366A6851745362436E642F6674386F54
505A53685845346B527338643272536B7151502B774968414D6241573249"
```

```
OK
AT+QESIM="trans",2878,0,2,78,"57496C6D6B576A6D6A585448422B5555646A4262665A636D34414F426174384558715450702270"
OK
```

2.4.3. Profile Download Modes

2.4.3.1. Internal Network Mode

In internal network mode, the module connects to the SM-DP+ server via HTTPS.

This mode applies to scenarios where the eSIM has a torrent profile and the module can use this profile to connect to the network.

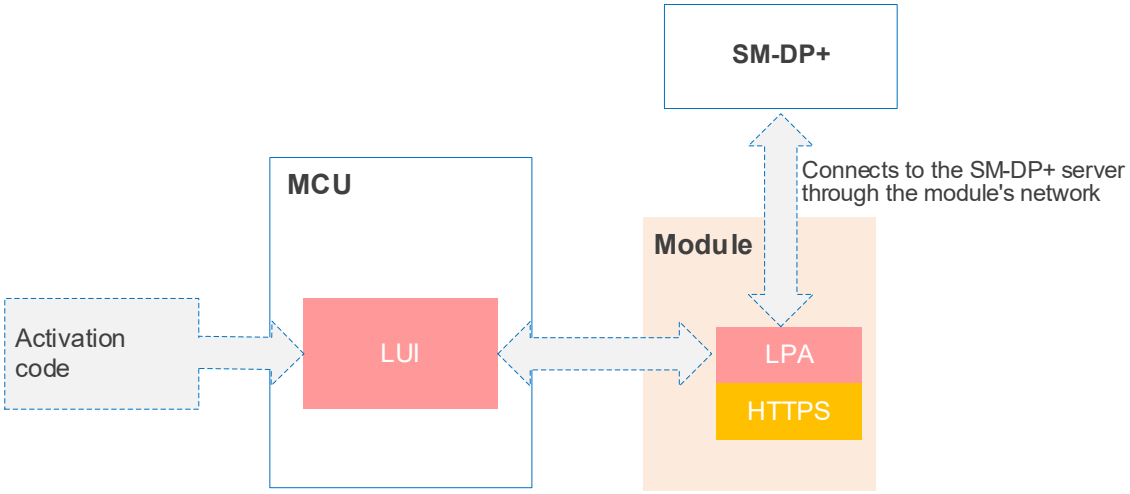


Figure 1: Internal Network Mode

2.4.3.2. External Network Mode

In external network mode:

- The MCU acts as a communication channel between the module and the SM-DP+ server for message transferring.
- The MCU connects to SM-DP+ via HTTPS, and connects to the module via UART interface.

This mode applies to scenarios where the eSIM does not have a torrent profile and the MCU has networking capability (e.g. Wi-Fi). See ***Figure 3*** for detailed profile download and installation process.

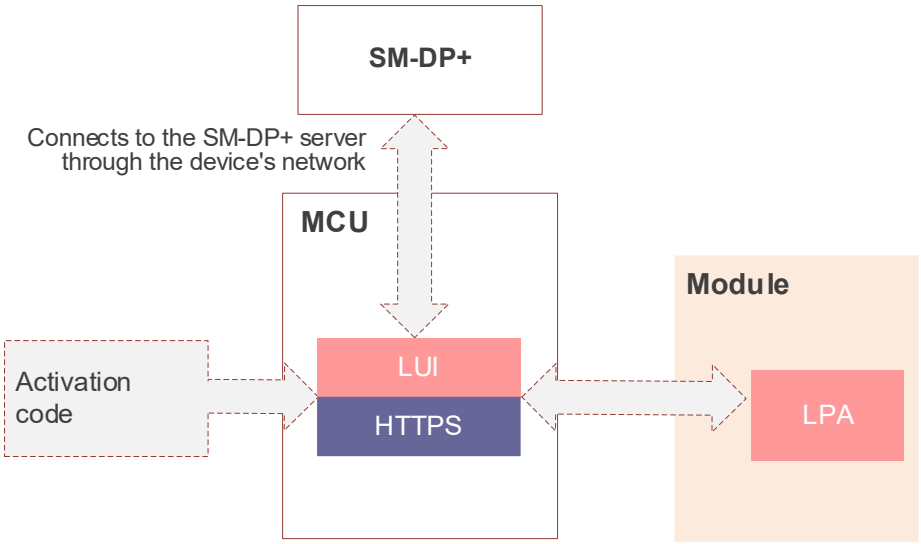


Figure 2: External Network Mode

2.4.3.3. Profile Download and Installation in External Network Mode

In external network mode, the MCU establishes a connection to the SM-DP+ server using HTTPS. The MCU sends data using the POST method. The URL and payload are obtained from the result codes reported by the module in the following format: **+QESIM: "trans",0,<URL>,<payloadlen>,<payload>**.

The HTTPS header content in a POST request is as follows:

Table 3: HTTPS Header Content in POST Request

Header	Value
X-Admin-Protocol	gsma/rsp/v2.2.0
Content-Type	application/json
charset	utf-8
User-Agent	gsma-rsp-lpad
Host	Host and port number, e.g. *demo-consumer.validspereachdpplus.com:443*
Content-Length	Content length

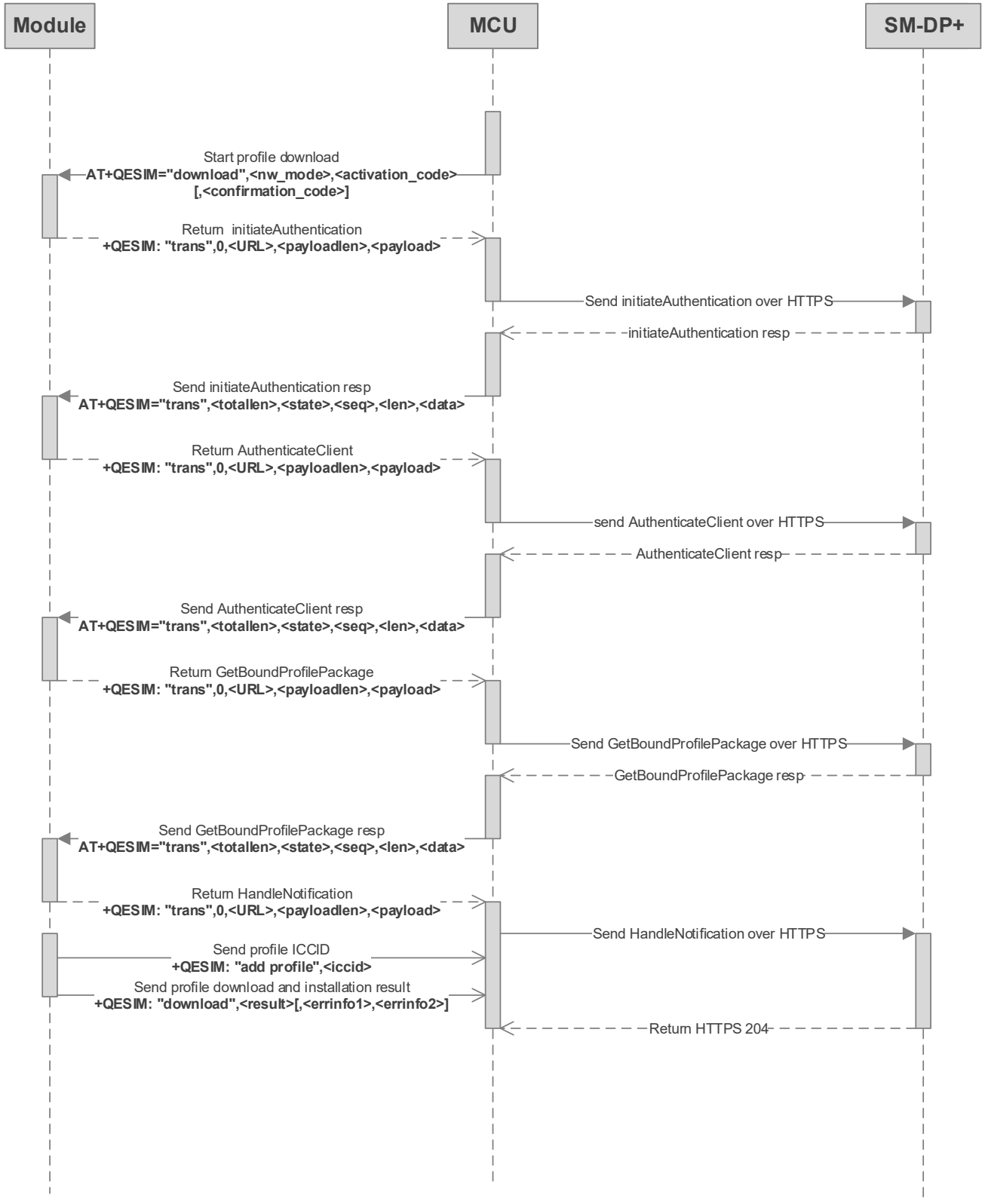


Figure 3: Flowchart for Profile Download and Installation in External Network Mode

2.5. eIM Service Commands

2.5.1. AT+QESIM="ipa_poll" Set Polling Cycle

This command sets the polling cycle.

AT+QESIM="ipa_poll" Set Polling Cycle	
Write Command **AT+QESIM="ipa_poll",<sec>**	Response **+QESIM: "ipa_poll",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configuration is saved automatically.

Parameter

<sec>	Integer type. Polling cycle, i.e. how often the module requests commands from eIM. 0 Module never requests commands from eIM 60–2147483647 Time interval at which the module requests commands from eIM. Default value: 600. Unit: seconds.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="ipa_poll",0          //Set the polling cycle to: Module never requests commands from eIM.
+QESIM: "ipa_poll",0

OK
AT+QESIM="ipa_poll",300        //Set the module polling cycle to 300 seconds.
+QESIM: "ipa_poll",0

OK
```

2.5.2. AT+QESIM="ipa_timer" Query Time Period Configuration and Status

This command queries the time period configuration and status.

AT+QESIM="ipa_timer" Query Time Period Configuration and Status	
Write Command **AT+QESIM="ipa_timer"**	Response **+QESIM: "ipa_timer",<result>,<poll_cycle>,<rollback>[,<fallback_flag>,<fallback_cycle>]** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	-

Parameter

<result>	Integer type. Result code. See ***Chapter 4*** for details.
<poll_cycle>	Integer type. Polling cycle, i.e. how often the module requests commands from eIM. 0 Module never requests commands from eIM 60–2147483647 Time interval at which the module requests commands from eIM Default value: 600. Unit: seconds.
<rollback>	Integer type. Rollback time limit. After profile switching, if the Internet cannot be accessed within **<rollback>**, the rollback function is triggered. Range: 0–2147483647. Default value: 300. Unit: seconds.
<fallback_flag>	Integer type. Whether to enable automatic executing of **AT+QESIM="fallback"** in case of continuous network abnormalities such as preferred network connection failure, extremely poor network quality and service outage. 0 Disable 1 Enable Parameter is not supported currently.
<fallback_cycle>	Integer type. Waiting time for automatic execution of **AT+QESIM="fallback"** in case of continuous network abnormalities. Range: 0–2147483647. Default value: 0 (indicates **AT+QESIM="fallback"** is not executed automatically). Unit: seconds. Parameter is not supported currently.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="ipa_timer"
+QESIM: "ipa_timer",0,600,300,0,0

OK
```

NOTE

<fallback_flag> and **<fallback_cycle>** are not supported currently, and they are always 0 in the response.

2.5.3. AT+QESIM="eim_add" Add First eIM Information

This command adds the first eIM information. You can execute this command successfully only if there is no eIM information in the module.

AT+QESIM="eim_add" Add First eIM Information	
Write Command **AT+QESIM="eim_add",<eIM_ID>,<eIM_FQDN>**	Response **+QESIM: "eim_add",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	This command takes effect immediately. The configuration is saved in eSIM.

Parameter

<eIM_ID>	String type. eIM ID.
<eIM_FQDN>	String type. eIM domain name.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="eim_add","1.3.6.1.4.1.58809.999.58809.1","iot-esim-global-gateway.quectel.com"
+QESIM: "eim_add",0
```

```
OK
```

2.5.4. AT+QESIM="eim_list" List Existing eIM Information

This command lists the existing eIM information.

AT+QESIM="eim_list" List Existing eIM Information	
Write Command **AT+QESIM="eim_list"**	Response **+QESIM: "eim_list",<result>** **[<eIM_eid>,<eIM_FQDN>,<eIM_type>** **[…]]** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	-

Parameter

<result>	Integer type. Result code. See ***Chapter 4*** for details.
<eIM_eid>	String type. eIM ID.
<eIM_FQDN>	String type. eIM domain name.
<eIM_type>	Integer type. eIM ID type. 1 OID 2 FQDN 3 Proprietary
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="eim_list"
+QESIM: "eim_list",0
"1.3.6.1.4.1.58809.999.58809.1","iot-esim-global-gateway.quectel.com",3

OK
```

2.5.5. AT+QESIM="eim_opt" Manage Existing eIM Information

This command manages existing eIM information.

AT+QESIM="eim_opt" Manage Existing eIM Information	
Write Command **AT+QESIM="eim_opt",<opt>,<eIM_ID>[,<eIM_FQDN>]**	Response **+QESIM: "eim_opt",<result>** **OK** If there is any error: **ERROR** Or **+CME ERROR: <err>**
Maximum Response Time	5 s
Characteristics	The command takes effect immediately. The configurations are saved automatically.

Parameter

<opt>	Integer type. Operation type. 0 Update **<eIM_FQDN>** information 1 Delete eIM information (The last eIM information cannot be deleted)
<eIM_ID>	String type. eIM ID.
<eIM_FQDN>	String type. eIM domain name. It can be omitted when **<opt>**=1.
<result>	Integer type. Result code. See ***Chapter 4*** for details.
<err>	Integer type. Error code. See ***document [1]*** for details.

Example

```
AT+QESIM="eim_opt",0,"1.3.6.1.4.1.58809.999.58809.1","iot-esim-global-gateway.quectel.com"
+QESIM: "eim_opt",0

OK
```

3 eSIM Related URCs

This chapter outlines the eSIM related URCs.

3.1. +QESIM: "download" Reports Profile Download and Installation Result

+QESIM: "download" Reports Profile Download and Installation Result	
+QESIM: "download",<result>[,<errinfo1>,<errinfo2>]	The URC reports the profile download and installation result. Upon successful profile download and installation, the related module resources are released.

Parameter

<result>	Integer type. Result code. See ***Chapter 4*** for details.
<errinfo1>	String type. Error information 1 (i.e. Subject Code). See Specific Status Codes tables in *SGP.22* for details.
<errinfo2>	String type. Error information 2 (i.e. Reason Code). See Specific Status Codes tables in *SGP.22* for details.

3.2. +QESIM: "add profile" Reports Profile ICCID

+QESIM: "add profile" Reports Profile ICCID	
+QESIM: "add profile",<iccid>	When a profile is successfully installed, this URC reports its corresponding ICCID.

Parameter

<iccid>	String type. Profile ICCID.

NOTE

The above URC applies only to the EC200U series and EG91xU family modules that use R01, R03 and R06 firmware versions. R01, R03 and R06 firmware refers to the firmware whose version number contains “R01”, “R03”, or “R06” as a segment, such as EC200UxxxxR01AxxMxx, EC200UxxxxR03AxxMxx, and EC200UxxxxR06AxxMxx. For details, please contact Quectel Technical Support.

3.3. +QESIM: "trans" Reports Profile Download Process and Result Information in External Network Mode

+QESIM: "trans" Reports Profile Download Process and Result Information in External Network Mode	
+QESIM: "trans",<result>,<URL>,<payloadlen>,<payload>	Reports the data to be sent to SM-DP+ during profile download process in external network mode. In this case **<result>** is 0.
+QESIM: "trans",<result>[,<errinfo1>,<errinfo2>]	Reports the error information for failed profile download or failed SM-DP+ message forwarding in external network mode. In this case **<result>** is not 0.

Parameter

<result>	Integer type. Result code. See ***Chapter 4*** for details.
<URL>	String type. HTTPS URL. This is the specific address used by SM-DP+ for message processing.
<payloadlen>	Integer type. Length of **<payload>**.
<payload>	String type. JASON format data to be sent to SM-DP+.
<errinfo1>	String type. Error information 1 (i.e. Subject Code). See Specific Status Codes tables in *SGP.22* for details.
<errinfo2>	String type. Error information 2 (i.e. Reason Code). See Specific Status Codes tables in *SGP.22* for details.

4 Summary of Result Codes

Table 4: Summary of Result Codes

<result>	Description
0	OK
85000001	Memory error
85000002	Invalid argument
85000003	Activation code error
85000004	Operation timeout
85000005	General error
85000006	Buffer overflow
85000007	Operation error
85000008	Message sending error
85000009	APDU sending error
8500000A	APDU status error
8500000B	APDU parsing error
8500000C	TLV parsing error
8500000D	JSON parsing error
8500000E	File system operation error
8500000F	HTTPS operation error
85000010	HTTPS busy error
85000011	System busy
85000012	Failed to get eUICC information 1
85000013	HTTPS message sending error

85000014	HTTPS response error
85000015	HTTPS head error
85000016	SMDP return error
85000017	Data waiting timeout
85000101	Failed to enable profile: ICCID or AID not found
85000102	Failed to enable profile: profile enabled already
85000103	Failed to enable profile: not allowed by policy
85000104	Failed to enable profile: profile re-enabling error
85000105	Failed to enable profile: CAT busy
8500017F	Failed to enable profile: undefined error
85000201	Failed to disable profile: ICCID or AID not found
85000202	Failed to disable profile: profile disabled already
85000203	Failed to disable profile: not allowed by policy
85000204	Failed to disable profile: CAT busy
8500027F	Failed to disable profile: undefined error
85000301	Failed to delete profile: ICCID or AID not found
85000302	Failed to delete profile: profile enabled already
85000303	Failed to delete profile: not allowed by policy
8500037F	Failed to delete profile: undefined error
85000401	Failed to list profile: incorrect input values
8500047F	Failed to list profile: undefined error
85000501	Failed to set nickname: ICCID not found
8500057F	Failed to set nickname: undefined error
85000601	Notification failure: nothing to delete
8500067F	Notification failure: undefined error
85000701	Authentication server error: invalid certificate

85000702	Authentication server error: invalid signature
85000703	Authentication server error: unsupported curve
85000704	Authentication server error: no session context
85000705	Authentication server error: invalid OID
85000706	Authentication server error: eUICC challenge mismatch
85000707	Authentication server error: CIPK unknown
8500077F	Authentication server error: undefined error
85000801	Download preparation error: invalid certificate
85000802	Download preparation error: invalid signature
85000803	Download preparation error: unsupported curve
85000804	Download preparation error: no session context
85000805	Download preparation error: invalid transaction ID
8500087F	Download preparation error: undefined error
85000901	Failed installation: incorrect input values
85000902	Failed installation: invalid signature
85000903	Failed installation: invalid transaction ID
85000904	Failed installation: unsupported CRT values
85000905	Failed installation: unsupported remote operation type
85000906	Failed installation: unsupported profile class
85000907	Failed installation: SECP03T structure error
85000908	Failed installation: SECP03T security error
85000909	Failed installation: ICCID already exists on eSIM
8500090A	Failed installation: insufficient memory for profile
8500090B	Failed installation: operation interruption
8500090C	Failed installation: PE processing error
8500090D	Failed installation: data mismatch

8500090E	Failed test profile installation: invalid NAA Key
8500090F	Failed installation: PPR not allowed
8500097F	Failed installation: undefined error
85000A02	Fallback profile error: profile enabled already
85000A05	Fallback profile error: CAT busy
85000A06	Fallback profile error: not available
85000A07	Fallback profile error: command error
85000A7F	Fallback profile error: undefined error

5 Appendix References

Table 5: Related Document

Document Name
[1] Quectel_EC200U&EG800G&EG91xU&EG915G_Series_AT_Commands_Manual

Table 6: Terms and Abbreviations

Abbreviation	Description
AID	Application Identifier
APDU	Application Protocol Data Unit
CAT	Card Application Toolkit
CIPK	Certificate Issuer Public Key
CRT	Control Reference Template
eIM	eSIM IoT Remote Manager
eSIM	Embedded Subscriber Identity Module
eUICC	Embedded Universal Integrated Circuit Card
FQDN	Fully Qualified Domain Name
GSMA	Global System for Mobile Communications Association
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICCID	Integrated Circuit Card Identifier
ID	Identifier
JSON	JavaScript Object Notation

LPA	Local Profile Assistant
LUI	Local User Interface
MCU	Microcontroller Unit
ME	Mobile Equipment
MOC	Mandatory, Optional or Conditional
NAA	Network Access Application
OID	Object Identifier
PE	Profile Element
PPR	Profile Policy Rules
QR	Quick Response
SIM	Subscriber Identity Module
SM-DP+	Subscription Manager Data Preparation+
TLV	Tag Length Value
UART	Universal Asynchronous Receiver/Transmitter.
URC	Unsolicited Result Code
URL	Uniform Resource Locator